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Are galaxies in compact groups special?

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ABSTRACT

We investigate the properties of galaxies in compact groups (CGs) and compare them to a control sample of galaxies.

Key words. galaxies: clusters: general – catalogs

1. Introduction

2. Data

2.1. Samples

We use the compact groups and control samples we built in our previous article Tricottet et al. (2025). We nevertheless briefly recall here how they were built. Explain. Contrarily to our previous article, though, we remove CGs that we classified as *split* following Zheng & Shen (2021). In this process, our initial compact group sample of 78 groups reduced to 62. We also remove groups bluit from Lim group 3688 from our Control$_{4B}$ and Control$_{4C}$ samples. This group has six galaxies having a redshift between 0.032 and 0.034, and one having a redshift of 0.048. This group seems spurious, and the outlier galaxy, altough not being the BGG, is selected both in our original Control$_{4B}$ and Control$_{4C}$, influencing the velocity dispersion of the group significantly (1979 and 1981 km s^{-1}, respectively).

We use the SDSS DR 16 to retrieve masses, star formation rates (SFRs) and morphologies of galaxies assessed in Galaxy Zoo Lintott et al. (2011). We specifically extracted fields `sfr_tot_p50`, `specsfr_tot_p50` and `lgm_tot_p50` from the `galSpecExtra` table, and `p_el_debiased` and `p_cs_debiased` from the `zooSpec` table.

2.2. sSFR

To build a general star formation classification that could be uniformly applied to all our samples, we extracted galaxies from the SDSS DR 16 through the query displayed in listing 1.

```
SELECT
    s.specObjID,
    s.z,
    p.petroMag_r,petroMag_u,petroMag_g,
        petroMag_i,petroMag_z,
    p.objID,
    g.sfr_tot_p50, g.specsfr_tot_p50, g.
        lgm_tot_p50,
    l.h_alpha_eqw, l.h_beta_eqw, l.
        oiii_5007_eqw, l.nii_6584_eqw,
    l.h_alpha_flux, l.h_beta_flux, l.
        oiii_5007_flux, l.nii_6584_flux,
    z. p_el_debiased AS p_E,
    z. p_cs_debiased    AS p_S
FROM SpecObj AS s
    JOIN PhotoObj AS p ON s.bestObjID = p
        .objID
    JOIN galSpecExtra as g ON s.specObjID
         = g.specObjID
    JOIN galSpecLine as l ON s.specObjID
        = l.specObjID
    JOIN zooSpec AS z ON s.specObjID = z.
        specObjID
WHERE s.z BETWEEN 0.005 AND 0.0452
    AND (p.petroMag_r - p.extinction_r <=
         17.77)
    AND s.class = 'GALAXY'
    AND g.lgm_tot_p50 > -1000
```

Listing 1: Query used for selecting spectral & photometric data from the SDSS

We select non-AGN galaxies by first placing them on the classical BPT diagnostic of Veilleux & Osterbrock (1987) and requiring measured values of both emission-line ratios

$$\log_{10}([\text{N II}]\lambda6584/\text{H}\alpha) \quad \text{and} \quad \log_{10}([\text{O III}]\lambda5007/\text{H}\beta).$$

A galaxy is flagged as an AGN if it satisfies either

$$\log_{10}\frac{[\text{N II}]}{\text{H}\alpha} > 0,$$

or if it lies above the empirical demarcation of Kauffmann et al. (2003):

$$\log_{10}\frac{[\text{O III}]}{\text{H}\beta} > \frac{0.61}{\log_{10}([\text{N II}]/\text{H}\alpha) - 0.05} + 1.3,$$

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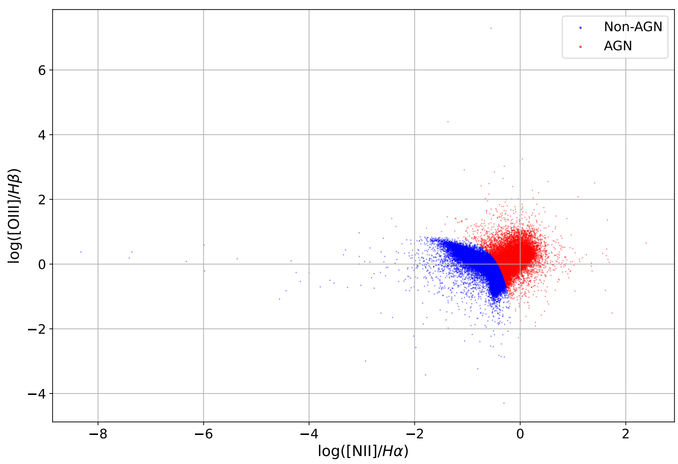

Fig. 1: BPT diagram showing our separation between ordinary and AGN galaxies

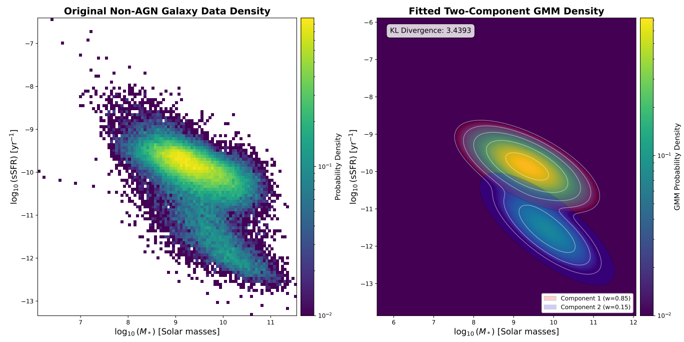

Fig. 2

in which case it is removed from the non-AGN sample. All remaining galaxies—including those with missing line ratios—are retained as non-AGN. The non-AGN selection process is shown on figure 1.

We identify the galaxies having a `specsf_tot_p50` of -9999 in the SDSS as quenched (for graphic representation purposes, we give it here an arbitrary low value of $10^{-14.0}$ Gyr^{-1}). We then consider the other ones and sort them between the star-forming and passive populations using a two-component Gaussian Mixture Model (GMM) in the $\log M_{*}$–$\log \mathrm{sSFR}$ plane. Its parameters determined by minimizing the Kullback–Leibler divergence between the empirical and model densities Kullback & Leibler (1951); McLachlan & Peel (2000). For any trial parameter vector $\boldsymbol{\theta}$, we reconstruct the mixture means $\{\boldsymbol{\mu}_i\}$, covariances $\{\boldsymbol{\Sigma}_i\}$, and weights $\{w_i\}$ via a Cholesky-like decomposition and logistic mapping, optionally constraining the second component's mean to isolate the passive population Dempster et al. (1977). The KL divergence is estimated by constructing a 2D histogram of the data and summing $p_{\mathrm{data}} \ln(p_{\mathrm{data}}/p_{\mathrm{GMM}})$, with a small regularization ϵ to avoid singularities Tojeiro et al. (2009); Bisigello et al. (2018). We optimize $\boldsymbol{\theta}$ via L-BFGS-B (with fallback to Nelder–Mead) over multiple guided and random initializations, selecting the solution with lowest divergence Pedregosa et al. (2011); Bishop (2006). Initial means are seeded by a median split in $\log \mathrm{sSFR}$ to improve convergence and robustness against local minima Rousseeuw (1987). In the final fit, one component naturally aligns with the high-sSFR "star-forming sequence," while the other captures the low-sSFR "passive" cloud, providing a data-driven division that agrees with previous multi-Gaussian decompositions of the SFR–M_{*} plane Wuyts et al. (2011); Hahn et al. (2019). The decision boundaries, defined as the loci where the posterior probabilities of adjacent components are equal, provide an objective criterion for delineating star-forming from passive galaxies.

Add relevant GM parameters found by the algorithm.

The sSFR status for each sample is shown in table 1. Tere is no significant differences of for Star forming proportion between CG_4 and the control samples (probabilities estimated using Fisher exact test are 6.5×10^{-1} versus

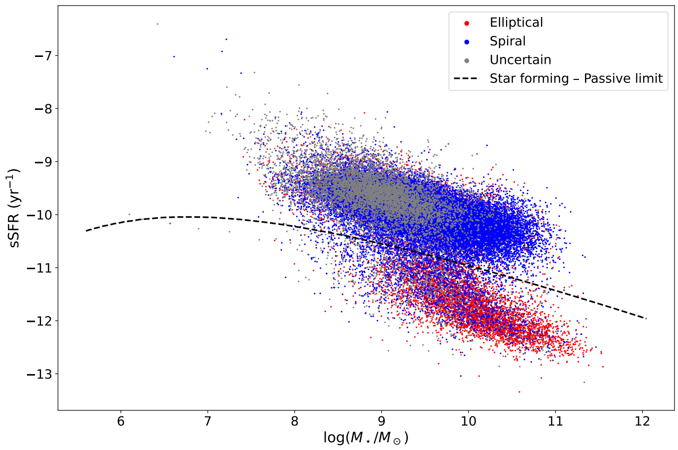

Fig. 3: sSFR vs mass, star-forming/passive limit and Zoo morphologies for the SDSS control sample, excluding quenched galaxies.

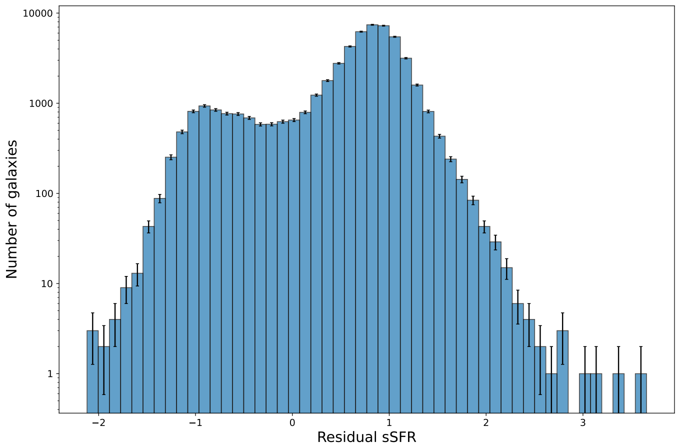

Fig. 4

$\mathrm{Control}_{4\mathrm{B}}$ sample, 2.5×10^{-1} versus $\mathrm{Control}_{4\mathrm{C}}$ and 1.8×10^{-2} versus RG_4).

Restrictions of the comparison to BGGs and satellites only are also shown in table 1. The p-values found from two-sided Fisher exact test are 0.58 against $\mathrm{Control}_{4\mathrm{B}}$, 0.59 against $\mathrm{Control}_{4\mathrm{C}}$ and 0.37 against RG_4. There is thus no significant difference in the fraction of star-forming BGGs nor satellites between CG_4 and our control samples either.

Sample		**Quenched**	**Passive**	**Starforming**
CG_4		18 (7.3 %)	146 (58.9 %)	84 (33.9 %)
	BGG	*6 (9.7 %)*	*45 (72.6 %)*	*11 (17.7 %)*
	Satellites	*12 (6.5 %)*	*101 (54.3 %)*	*73 (39.2 %)*
$Control_{4B}$		181 (6.5 %)	1596 (57.1 %)	1019 (36.4 %)
	BGG	*58 (8.3 %)*	*534 (76.4 %)*	*107 (15.3 %)*
	Satellites	*123 (5.9 %)*	*1062 (50.6 %)*	*912 (43.5 %)*
$Control_{4C}$		191 (6.3 %)	1619 (53.8 %)	1198 (39.8 %)
	BGG	*65 (8.6 %)*	*569 (75.7 %)*	*118 (15.7 %)*
	Satellites	*126 (5.6 %)*	*1050 (46.5 %)*	*1080 (47.9 %)*
RG_4		5 (2.2 %)	101 (45.1 %)	118 (52.7 %)
	BGG	*3 (5.4 %)*	*38 (67.9 %)*	*15 (26.8 %)*
	Satellites	*2 (1.2 %)*	*63 (37.5 %)*	*103 (61.3 %)*
SDSS selection		*634 (1.2 %)*	*7705 (14.7 %)*	*44192 (84.1 %)*

Table 1: Number of galaxies in each sSFR status for each sample.

2.3. Morphology

Morphologies are determined via the Galaxy Zoo citizen-science decision tree, in which each galaxy image receives multiple independent volunteer classifications that are aggregated into debiaised vote fractions Lintott et al. (2011). Those fractions are provided in the `zooSpecz` table of the SDSS databasethrough the `p_el_debiased` and `p_cs_debiased` columns. We attribute to galaxies the morphology "elliptical" if the `p_el_debiased` value is greater than 0.5, and "spiral" if the `p_cs_debiased` value is greater than 0.5. Other cases are considered as "uncertain". The morphologies found for each sample are shown in table 2. Barnard two-sided exact tests were performed to compare the fraction of each morphology between CG_4 and each control sample. The p-values found are for ellipticals and for spirals versus $Control_{4B}$ sample, and versus $Control_{4C}$ and and versus RG_4. Morphologies of CG_4, wich we identified as cores of richer groups in Tricottet et al. (2025), have morphologies significantly different from ordinary groups of four members. Figure 5 displays the sSFR against masses for galaxies from the SDSS selection and for CG_4s galaxies.

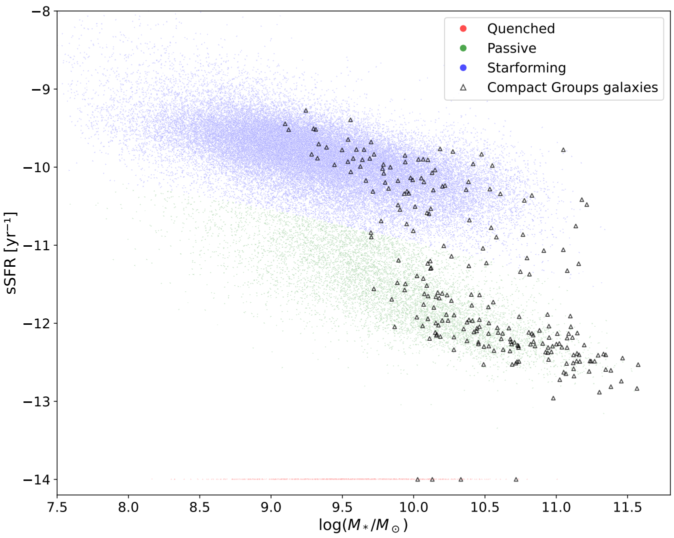

Fig. 5: sSFR and morphologies of galaxies in CG_4 and our SDSS selection. Quenched galaxies are represented at an arbitrary low value of $10^{-14.0}$ Gyr^{-1}.

2.4. Morphology vs sSFR

Starforming CG_4 are composed of of their galaxies classified as spiral (% once galaxies having uncertain morphology removed) and (% once galaxies having uncertain morphology removed) elliptical. These figures are (%) spiral and (%) elliptical galaxies for $Control_{4B}$, (%) spiral and (%) elliptical galaxies for $Control_{4C}$ and (%) spiral and (%) elliptical galaxies for RG_4. The probability that those fractions come from the same distribution is estimated from the as against $Control_{4B}$, against $Control_{4C}$ and against RG_4. There is thus neither significant excess nor lack of starforming elliptical galaxies in CG_4 compared to all our control samples.

If we restrict the comparison to BGGs only, we find Elliptical BGGs (% of the BGGs) and Spiral BGGs (%) in CG_4. This is to be compared to (%) Elliptical and (%) Spiral galaxies in $Control_{4B}$, (%) Elliptical and (%) Spiral in $Control_{4C}$, and (%) Elliptical and (%) Quenched in RG_4. The p-values found from are against $Control_{4B}$, against $Control_{4C}$ and against RG_4. There is thus no significant difference in the fraction of star-forming BGGs between CG_4 and our control samples.

3. sSFR

3.1. Main sequence

We model the star-forming main sequence as a power law in stellar mass, to various orders (1 to 4) - cf figure 9. We then choose to use order 2 to compute the residuals of each star-forming galaxy to the main sequence, and plot their histograms in figure 10.

The median main-sequence residual of CG4 galaxies is significantly different

from that of Control4B, with a median offset of $\Delta = 0.056$ ($0.027 \leq \Delta \leq 0.098$, bootstrap), corresponding to a p-value of $p = 0.055$.

The median main-sequence residual of CG4 galaxies is significantly different

from that of Control4C, with a median offset of $\Delta = 0.082$ ($0.062 \leq \Delta \leq 0.122$, bootstrap), corresponding to a p-value of $p = 0.008$.

The median main-sequence residual of CG4 galaxies is not significantly different

Sample	**Elliptical**	**Spiral**	**Uncertain**
CG_4	124 (%)	86 (%)	38 (%)
$Control_{4B}$	1107 (%)	1268 (%)	421 (%)
$Control_{4C}$	1215 (%)	1251 (%)	542 (%)
RG_4	60 (%)	129 (%)	35 (%)
SDSS selection	*10091 (%)*	*34715 (%)*	*7725 (%)*

Table 2: Number of galaxies of each morphology for each sample.

Sample	$\langle \log sSFR \rangle$		$\langle M_r \rangle$		$\langle \log(M_\star/M_\odot) \rangle$	
	Passive	Star-forming	Passive	Star-forming	Passive	Star-forming
CG4	−12.08	−10.19	−21.16	−20.4	10.64	10.05
Control4B	−12.1	−10.31	−21.52	−20.75	10.79	10.24
Control4C	−11.92	−10.16	−20.81	−19.93	10.47	9.82
RG4	−11.91	−10.16	−21.11	−20.32	10.62	9.97

Table 3: Median values of galaxy properties for passive and star-forming populations in each sample. Error bars are computed via bootstrapping (10 000 iterations).

Sample	sSFR (yr^{-1})		M_r		$\log(M_\star/M_\odot)$	
	Passive BGG	Star-forming BGG	Passive BGG	Star-forming BGG	Passive BGG	Star-forming BGG
CG4	-11.53 ± 0.09	-10.85 ± 0.29	-20.93 ± 0.11	-20.77 ± 0.32	10.46 ± 0.06	10.32 ± 0.12
Control4B	-11.53 ± 0.03	-10.86 ± 0.06	-21.25 ± 0.03	-21.12 ± 0.07	10.61 ± 0.02	10.46 ± 0.04
Control4C	-11.3 ± 0.03	-10.62 ± 0.06	-20.42 ± 0.05	-20.55 ± 0.08	10.22 ± 0.02	10.15 ± 0.06
RG4	-11.14 ± 0.19	-10.58 ± 0.16	-20.69 ± 0.14	-20.74 ± 0.28	10.32 ± 0.07	10.19 ± 0.13

Table 4: Median values of various quantities for galaxies according to the fertility of their BGG (BGG included). Error bars are computed via bootstrapping (10 000 iterations).

from that of RG4, with a median offset of $\Delta = 0.046$ ($-0.036 \leq \Delta \leq 0.098$, bootstrap), corresponding to a p-value of $p = 0.535$.

A direct rank-sum comparison of the star-forming main-sequence offsets only identifies significant differences between CG_4 and

$Control_{4C}$, whose median offset is -0.060 with respect to the fitted sequence (p = 0.03).

Within CG_4, the star-forming main-sequence offsets of the Embedded, Isolated and Predom classes remain statistically consistent with one another.

3.2. Distance to the BGG

We also compare the sSFR of satellite galaxies with their distance to the BGG, expressed both in projected kpc and in units of $\langle R_{ij} \rangle$.

Significant monotonic trends are detected in the following cases.

In CG_4, sSFR correlates with projected distance to the BGG with Spearman coefficient $\rho_S = 0.24$ for 177 satellites ($p = 1.2 \times 10^{-3}$) and Pearson coefficient $r = 0.21$ ($p = 4.1 \times 10^{-3}$).

In $Control_{4B}$, sSFR correlates with distance to the BGG normalized by $\langle R_{ij} \rangle$ with Spearman coefficient $\rho_S = 0.08$ for 1991 satellites ($p = 3.6 \times 10^{-4}$) and Pearson coefficient $r = 0.08$ ($p = 2.3 \times 10^{-4}$).

In $Control_{4C}$, sSFR correlates with projected distance to the BGG with Spearman coefficient $\rho_S = 0.25$ for 2143 satellites ($p = < 10^{-6}$) and Pearson coefficient $r = 0.20$ ($p = < 10^{-6}$).

In $Control_{4C}$, sSFR correlates with distance to the BGG normalized by $\langle R_{ij} \rangle$ with Spearman coefficient $\rho_S = 0.10$ for 2143 satellites ($p = 4.6 \times 10^{-6}$) and Pearson coefficient $r = 0.09$ ($p = 1.8 \times 10^{-5}$).

3.3. Dominance

We split groups into two categories: dominated groups, for which the BGG luminosity fraction exceeds our threshold of 0.6, and non-dominated groups. Among the group properties explored in this work, the quantities showing at least one significant dominated/non-dominated difference are

$\left\langle R_{ij} \right\rangle$ (kpc),
$\Delta_{\rm BGG-cen}/\langle R_{ij} \rangle$,
σ_v (km s^{-1}),
$\Delta V_{\rm BGG}/\sigma_v$ and
$L_{\rm group}$ ($L_\odot$).

In CG_4, no dominated/non-dominated difference remains significant after the FDR correction.

In $Control_{4B}$, the strongest corrected dominated/non-dominated separation concerns $L_{\rm group}$ ($L_\odot$): the dominated groups have a median value of 9.1e+10, compared to 1.28e+11 for the non-dominated ones ($p_{\rm adj} = < 10^{-6}$; Cliff's δ = -0.55).

In $Control_{4C}$, the strongest corrected dominated/non-dominated separation concerns $L_{\rm group}$ ($L_\odot$): the dominated groups have a median value of 8.49e+10, compared to 9.54e+10 for the non-dominated ones ($p_{\rm adj} = < 10^{-6}$; Cliff's δ = -0.25).

Sample	sSFR (yr^{-1})		M_r		$\log(M_\star/M_\odot)$	
	Passive BGG	Star-forming BGG	Passive BGG	Star-forming BGG	Passive BGG	Star-forming BGG
CG4	−11.23 ± 0.16	−10.92 ± 0.48	−20.45 ± 0.13	−20.34 ± 0.18	10.22 ± 0.05	10.09 ± 0.06
Control4B	−11.26 ± 0.06	−10.91 ± 0.13	−20.9 ± 0.03	−20.75 ± 0.06	10.43 ± 0.02	10.3 ± 0.03
Control4C	−10.96 ± 0.04	−10.6 ± 0.07	−19.8 ± 0.04	−20.02 ± 0.10	9.91 ± 0.02	9.89 ± 0.07
RG4	−10.82 ± 0.19	−10.61 ± 0.22	−20.23 ± 0.17	−20.3 ± 0.13	10.09 ± 0.08	10 ± 0.13

Table 5: Median values of various quantities for galaxies according to the fertility of their BGG (satellites only). Error bars are computed via bootstrapping (10 000 iterations).

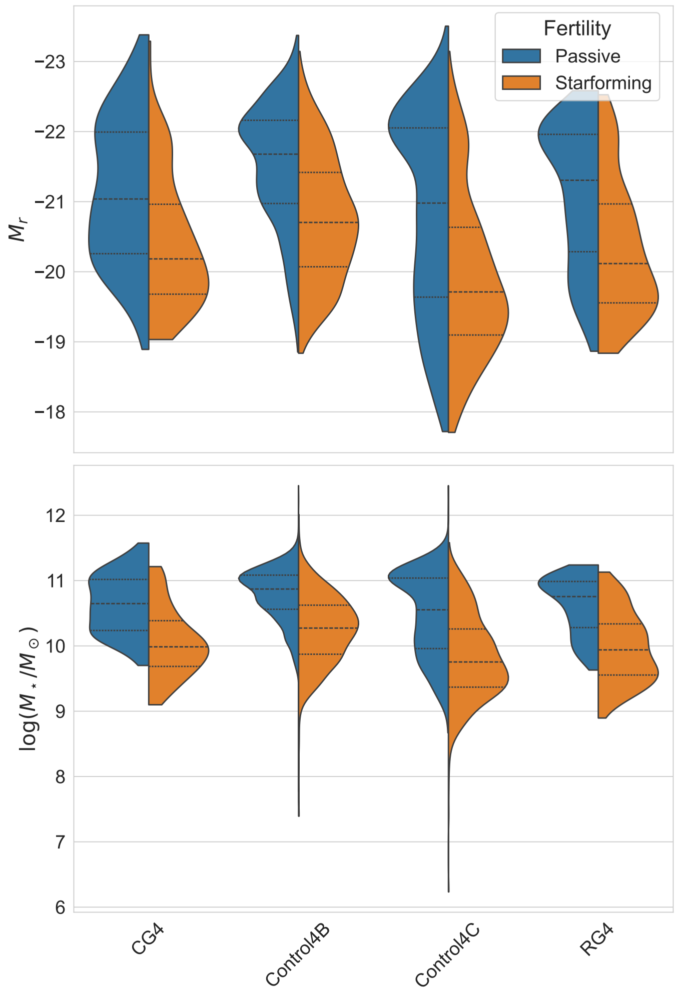

Fig. 6: Distribution various quantities for galaxies according their fertility.

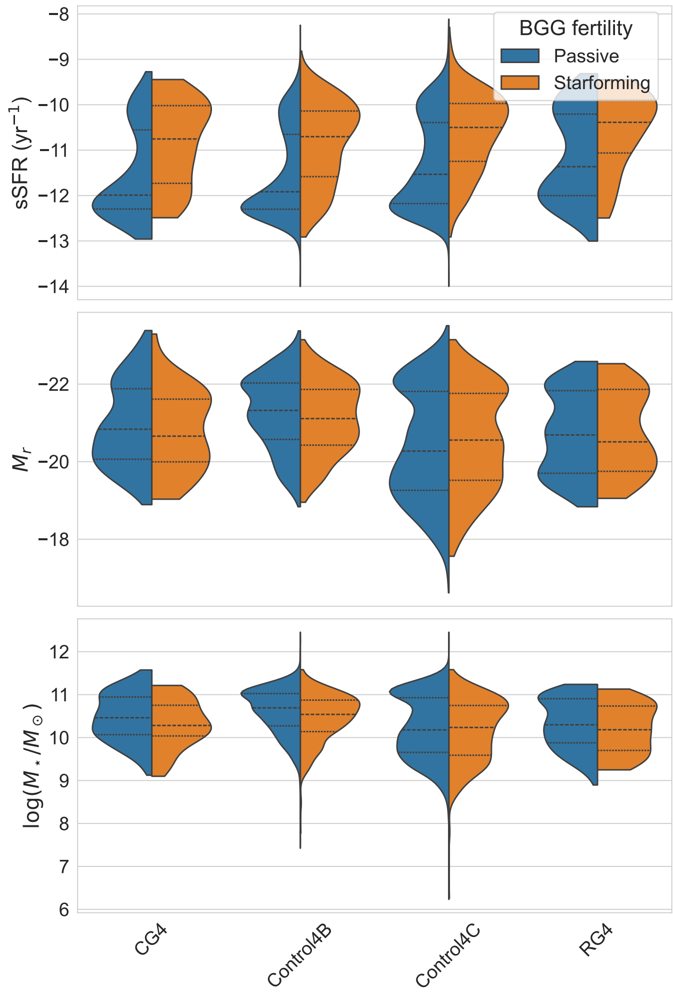

Fig. 7: Distribution various quantities for galaxies according to the fertility of their BGG (BGG included).

In RG$_4$, the strongest corrected dominated/non-dominated separation concerns $L_{\rm group}$ ($L_\odot$): the dominated groups have a median value of 7.09e+10, compared to 9.82e+10 for the non-dominated ones ($p_{\rm adj}$ = 7.8 × 10^{-5}; Cliff's δ = -0.59).

The group spiral fraction also correlates significantly with the group-scale observables in a subset of the samples.

For Control$_{4C}$ dominated groups, $S_{\rm frac}$ correlates with $\langle R_{ij} \rangle$ (kpc) ($\rho_{\rm S}$ = 0.38, 527 groups, p = < 10^{-6}).

For Control$_{4C}$ non-dominated groups, $S_{\rm frac}$ correlates with $\langle R_{ij} \rangle$ (kpc) ($\rho_{\rm S}$ = 0.35, 224 groups, p = < 10^{-6}).

For Control$_{4C}$ dominated groups, $S_{\rm frac}$ correlates with σ_v (km s^{-1}) ($\rho_{\rm S}$ = −0.23, 527 groups, p = < 10^{-6}).

For Control$_{4C}$ dominated groups, $S_{\rm frac}$ correlates with $L_{\rm group}$ ($L_\odot$) ($\rho_{\rm S}$ = −0.15, 527 groups, p = 8.4×10^{-4}).

For Control$_{4B}$ non-dominated groups, $S_{\rm frac}$ correlates with $L_{\rm group}$ ($L_\odot$) ($\rho_{\rm S}$ = −0.14, 504 groups, p = 1.4×10^{-3}).

For CG$_4$ dominated groups, $S_{\rm frac}$ correlates with $L_{\rm group}$ ($L_\odot$) ($\rho_{\rm S}$ = −0.50, 36 groups, p = 1.9×10^{-3}).

For Control$_{4C}$ non-dominated groups, $S_{\rm frac}$ correlates with σ_v (km s^{-1}) ($\rho_{\rm S}$ = −0.20, 224 groups, p = 2.6×10^{-3}).

For CG$_4$ non-dominated groups, $S_{\rm frac}$ correlates with $\Delta V_{\rm BGG}/\sigma_v$ ($\rho_{\rm S}$ = −0.53, 26 groups, p = 5.6×10^{-3}).

For Control$_{4C}$ non-dominated groups, $S_{\rm frac}$ correlates with $L_{\rm group}$ ($L_\odot$) ($\rho_{\rm S}$ = −0.15, 224 groups, p = 0.02).

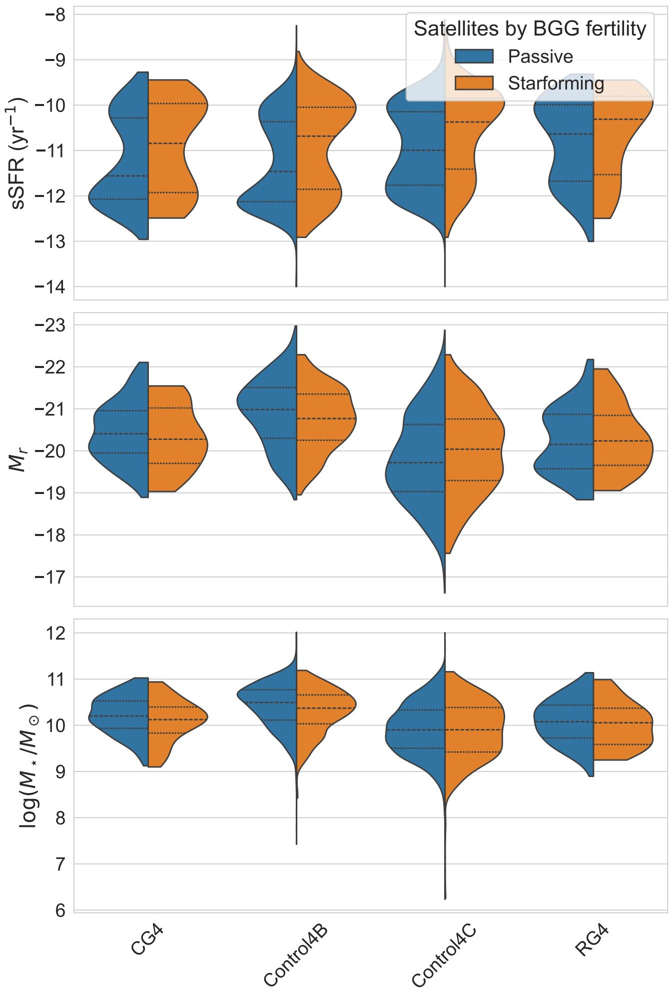

Fig. 8: Distribution various quantities for galaxies according to the fertility of their BGG (BGG excluded).

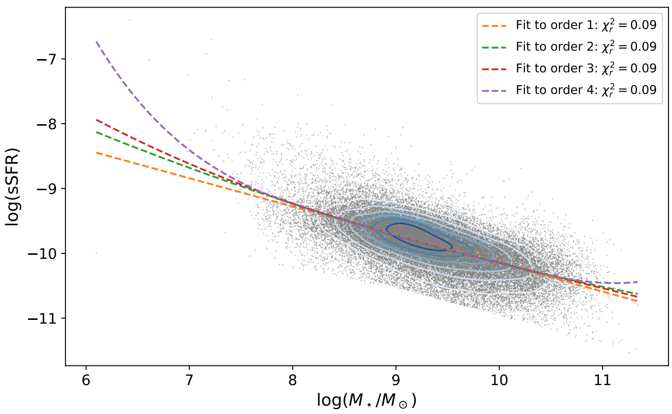

Fig. 9

For Control$_{4B}$ non-dominated groups, $S_{\rm frac}$ correlates with σ_v (km s^{-1}) ($\rho_S = -0.09$, 504 groups, p = 0.03).

For Control$_{4B}$ non-dominated groups, $S_{\rm frac}$ correlates with $\langle R_{ij} \rangle$ (kpc) ($\rho_S = 0.09$, 504 groups, p = 0.04).

For CG$_4$ non-dominated groups, $S_{\rm frac}$ correlates with $\langle R_{ij} \rangle$ (kpc) ($\rho_S = 0.40$, 26 groups, p = 0.04).

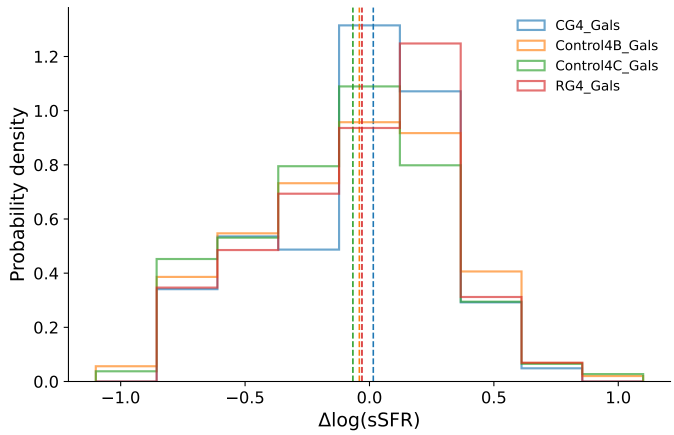

Fig. 10

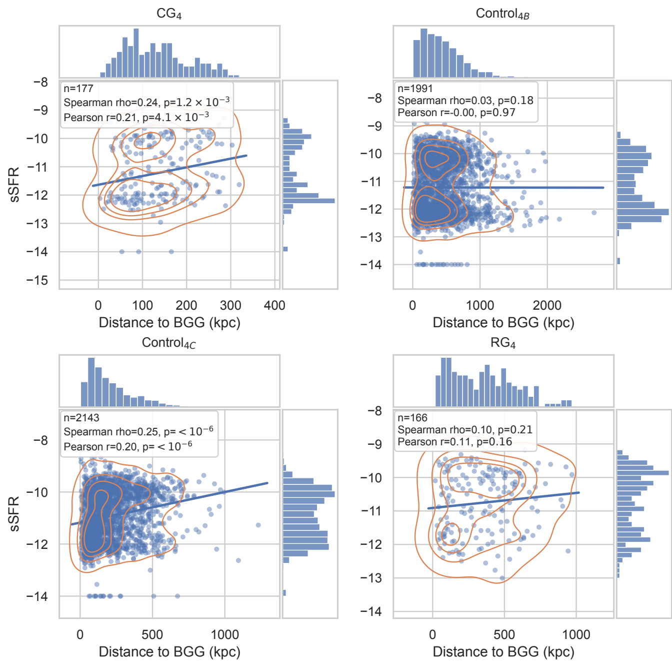

Fig. 11: sSFR as a function of projected distance to the BGG, in kpc, for satellite galaxies in each sample.

3.4. Morphology and domination

When we further split the samples by domination and by the morphology of the BGG, significant effects only appear in

Control$_{4B}$ dominated groups, where the median satellite spiral fraction is 1.00 for spiral BGGs and 0.67 for elliptical BGGs. The corresponding group-level odds ratio is 1.84 (p = 0.01), while the satellite-level clustered fit gives an odds ratio of 1.84 (p = 0.02);

Control$_{4C}$ dominated groups, where the median satellite spiral fraction is 0.67 for spiral BGGs and 0.50 for elliptical BGGs. The corresponding group-level odds ratio is 1.84 (p = 2.1×10^{-5}), while the satellite-level clustered fit gives an odds ratio of 1.84 (p = 2.7×10^{-5}).

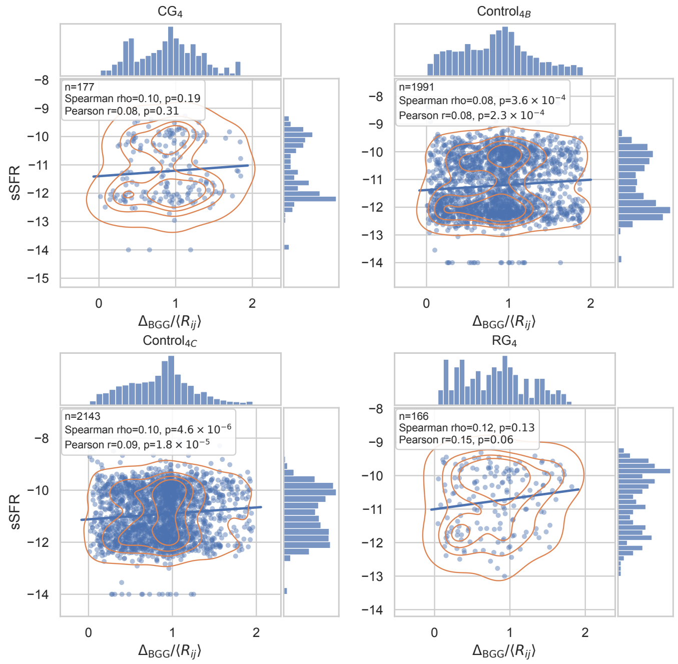

Fig. 12: sSFR as a function of projected distance to the BGG normalized by $\langle R_{ij} \rangle$ for satellite galaxies in each sample.

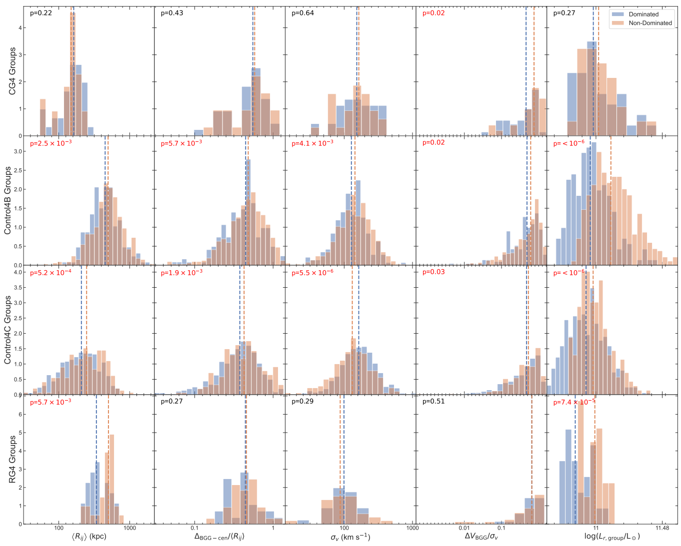

Fig. 13: Distributions of the main group-scale observables in dominated and non-dominated systems.

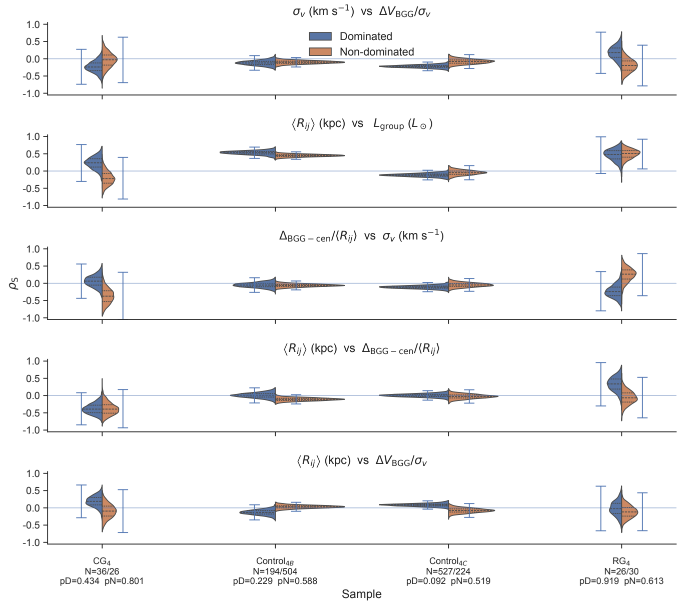

Fig. 14: Bootstrap distributions of the Spearman coefficients for the group-property pairs selected by the XOR domination criterion.

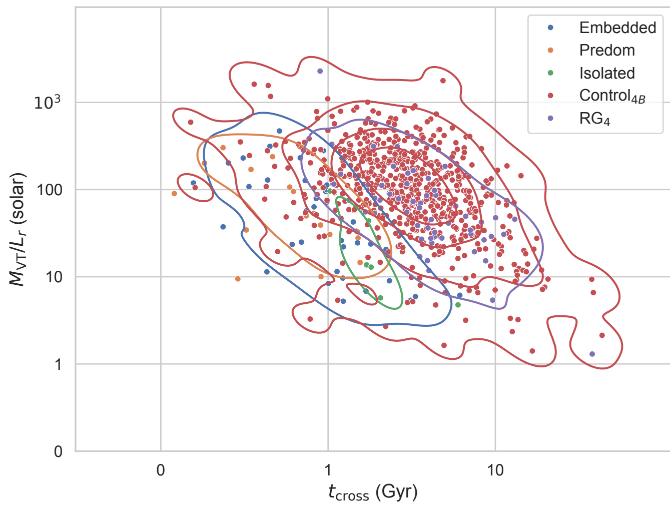

Fig. 15: Crossing time versus virial mass-to-light ratio for the compact groups and their controls.

4. Group Dynamics

The crossing-time exploration quantifies the trends illustrated in figures 15 to 17.

For $\log(\mathcal{M}_{\rm VT}/L_r)$, the global Spearman coefficient with $\log t_{\rm cross}$ is $\rho_{\rm S} = -0.39$ based on 816 groups (p = $< 10^{-6}$).

For $\log(\mathcal{M}_{\rm VT})$, the global Spearman coefficient with $\log t_{\rm cross}$ is $\rho_{\rm S} = -0.33$ based on 816 groups (p = $< 10^{-6}$).

For $\log(L_{\rm group})$, the global Spearman coefficient with $\log t_{\rm cross}$ is $\rho_{\rm S} = 0.08$ based on 816 groups (p = 0.03).

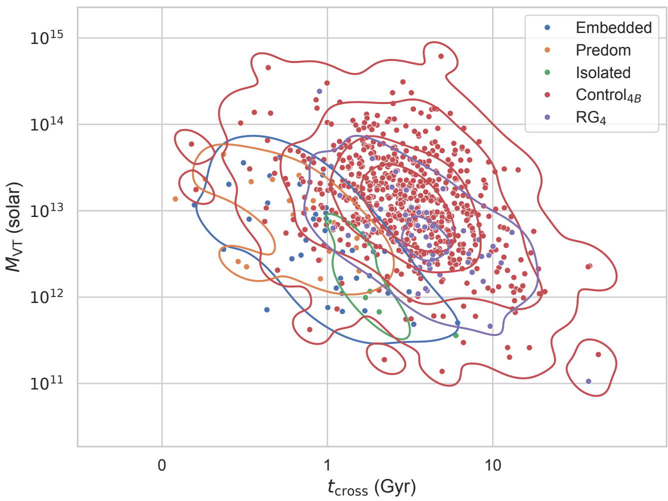

Fig. 16: Crossing time versus virial mass for the compact groups and their controls.

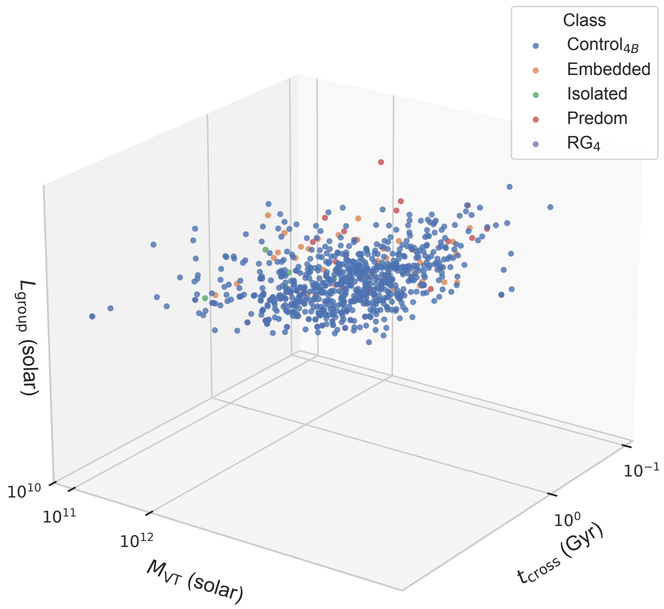

Fig. 17: Three-dimensional view of the crossing time, virial mass and group luminosity space.

Interpolation points for sSFR classification

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References

Hahn, C., Starkenburg, T. K., Choi, E., et al. 2019, Astrophys. J., 872
Wuyts, S., Förster Schreiber, N. M., Lutz, D., et al. 2011, Astrophys. J., 742, 96
Zheng, Y.-L. & Shen, S.-Y. 2021, ApJ, 911, 105
Bishop, C. M. 2006, Pattern Recognition and Machine Learning (Springer)
Tojeiro, R., Heavens, A. F., Jimenez, R., & Panter, B. 2009, Mon. Not. R. Astron. Soc., 397, 1263
Lintott, C., Schawinski, K., Bamford, S., et al. 2011, MNRAS, 410, 166
Kullback, S. & Leibler, R. A. 1951, Ann. Math. Statist., 22, 79
Veilleux, S. & Osterbrock, D. E. 1987, Astrophysical Journal Supplement Series, 63, 295
Dempster, A. P., Laird, N. M., & Rubin, D. B. 1977, J. R. Statist. Soc. B, 39, 1
McLachlan, G. & Peel, D. 2000, Finite Mixture Models (Wiley)
Tricottet, M., Mamon, G. A., & Díaz-Giménez, E. 2025, A&A, 699, A329
Bisigello, L., Caputi, K. I., Grogin, N., & Koekemoer, A. 2018, Astron. Astrophys., 609, A82
Kauffmann, G., Heckman, T. M., Tremonti, C., et al. 2003, Monthly Notices of the Royal Astronomical Society, 346, 1055
Pedregosa, F., Varoquaux, G., Gramfort, A., et al. 2011, J. Mach. Learn. Res., 12, 2825
Rousseeuw, P. J. 1987, Comput. Appl. Math., 20, 53